

Boosting Facial Emotion Recognition with Attention-Enhanced Residual Networks

Integrating CBAM into Residual Masking to Strengthen Feature Learning

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BACKGROUND

Facial emotion recognition can improve autism therapy, audience reaction analysis, expressive video communication, and psychiatric evaluations. Existing models often miss subtle emotions or overfit noisy datasets like FER-2013. Our model enhances key facial features while filtering noise, leading to more accurate and real-world ready emotion recognition.

METHODS

Dataset Preparation: Collected facial emotion images from FER-2013 (~35k grayscale images) and RAF-DB (~15k real world images), covering seven emotions, all images were standardized at 48×48 resolution. From the combined training data, a 90%-10% split was applied, resulting in 36,882 images for training and 4,098 images for validation. The official combined test sets, totaling 10,246 images, were kept separate for final evaluation.

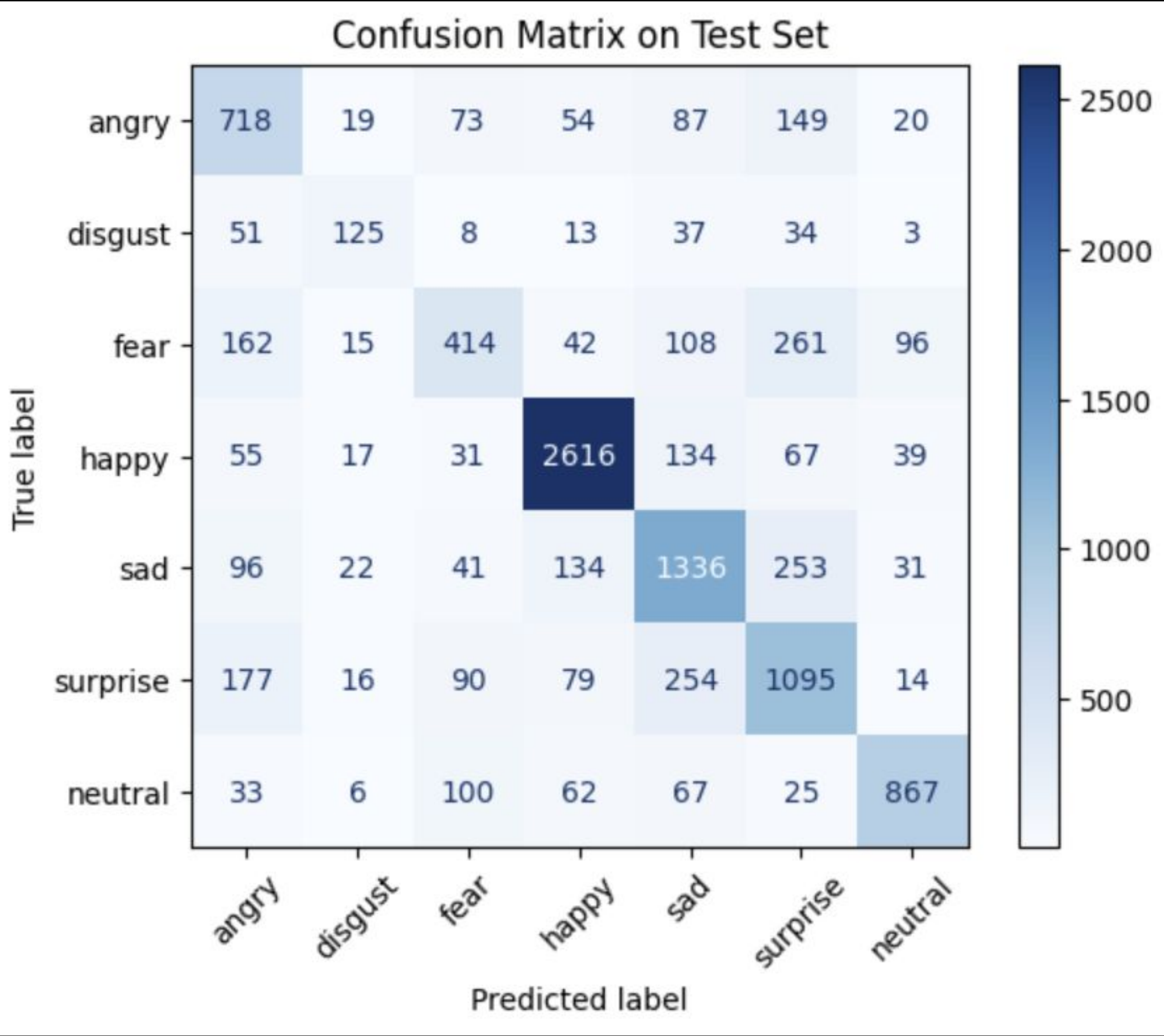
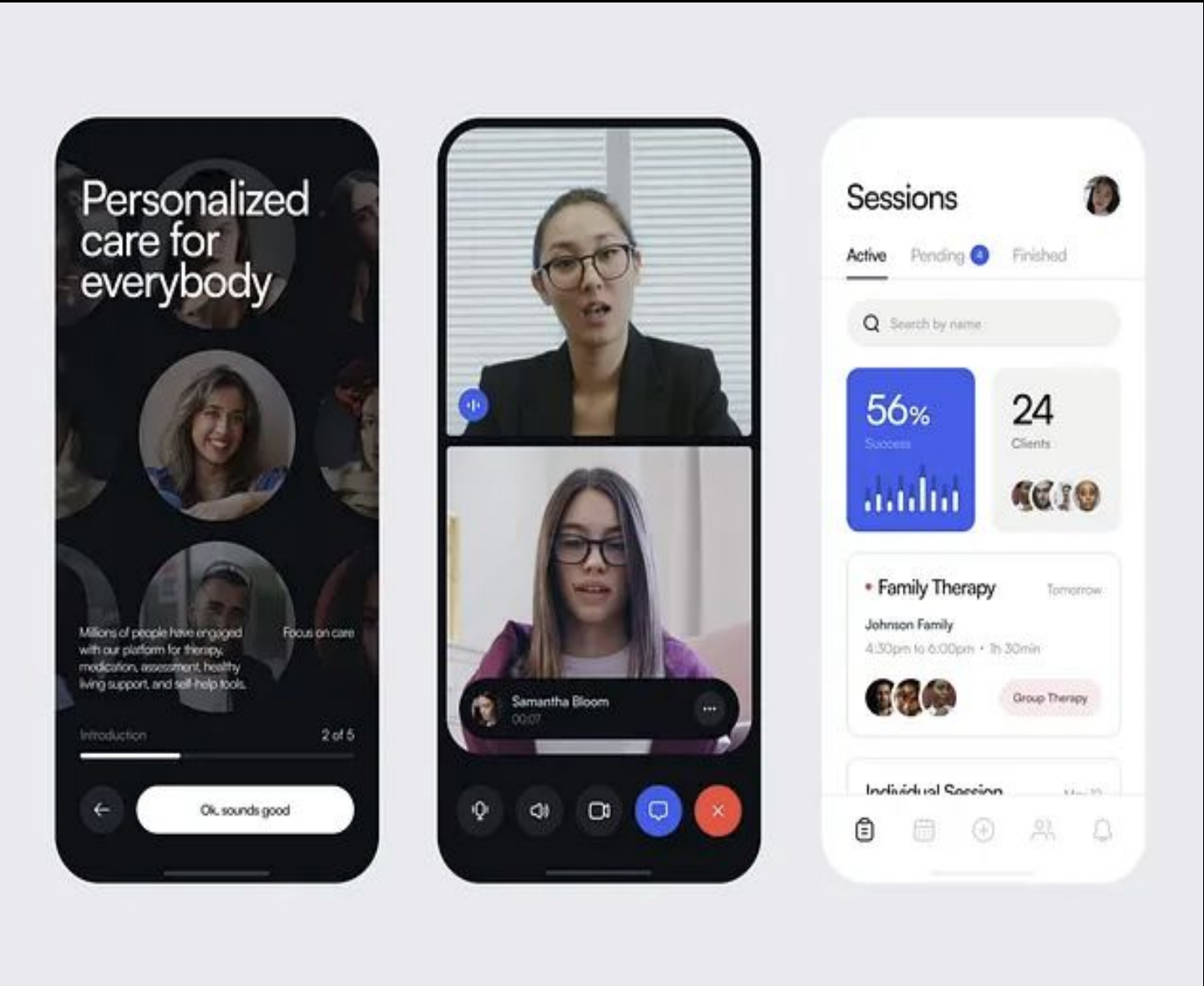
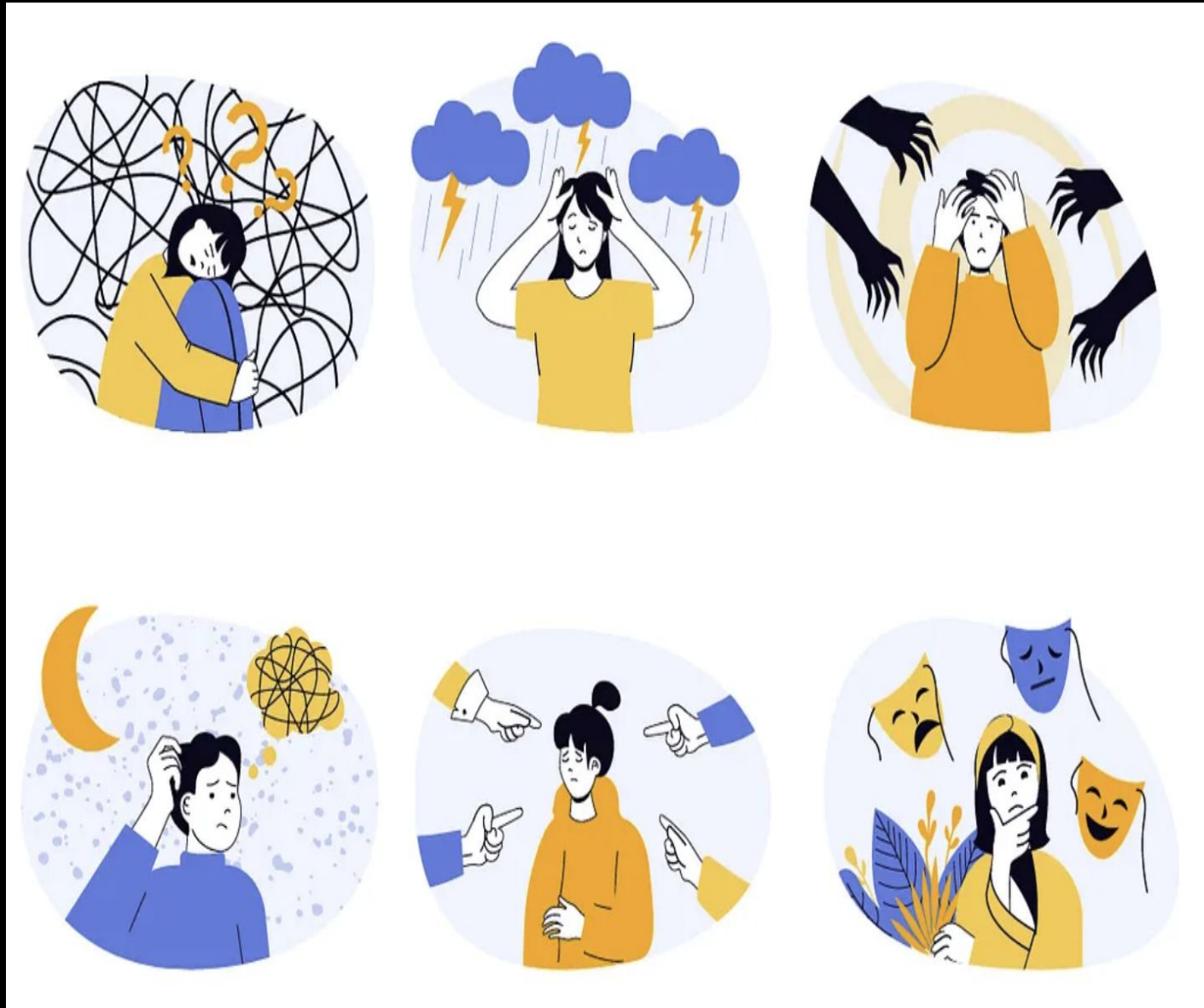
Model Architecture and Training Setup: The model consists of three convolutional blocks, each composed of Conv2D→BatchNorm→ReLU→Residual Mask Block→CBAM→Max Pooling. Residual Masking introduces residual connections to improve feature propagation. Whereas, CBAM helps for better spatial and channel-wise feature refinement. Finally, A dropout layer (rate = 0.4) is applied before a fully connected layer that outputs predictions across seven emotion classes.

- Loss Function: Cross Entropy Loss
- Optimizer: Adam (learning rate 0.0005, weight decay 1e-5)
- Epochs: 30, Batch Size: 64
- Learning Rate Scheduler: StepLR (decay factor 0.5 every 10 epochs)

RESULTS

- We evaluated the performance of our (CNN + RMN + CBAM) model using key metrics including accuracy, precision, recall, and F1-score.
- The model achieved a final validation accuracy of 69.64% and a test accuracy of 70.69%.
- On the test set, it obtained a macro-averaged precision of 66.08%, recall of 63.51%, and a macro F1-score of 65.11%, indicating balanced performance across all emotion classes.

Attention-enhanced Residual Masking, combined with careful hyperparameter tuning, improved feature focus, reduced overfitting, and boosted generalization on noisy datasets, enabling accurate detection of subtle emotions in a lightweight model.



Model	Model Description	Final Val Acc	Final Test Acc	Macro F1 Score
Model 1	CNN	65.64%	66.30%	0.61
Model 2	CNN + RMN	68.79%	70.44%	0.63
Model 3	CNN + RMN + CBAM	69.64%	70.69%	0.65

Class	Precision (%)	Recall (%)	F1-Score (%)	Support
angry	56.66	64.55	60.35	1120
disgust	60.49	45.76	52.10	271
fear	54.85	38.62	45.32	1098
happy	88.31	88.31	88.31	2959
sad	64.45	69.26	66.77	1913
surprise	56.71	62.20	59.33	1725
neutral	81.11	75.86	78.40	1160

Classification Matrix: High F1-scores were observed for 'happy' and 'neutral', while 'fear' and 'disgust' were more challenging, reflecting the impact of class imbalance on our (CNN + RMN + CBAM) model performance.

